

MSTAR PRACTICE WORKSHEET

Lesson 3-4: Determine Equivalent Ratios Using Graphs

Grade 6 Mathematics · Standard 6.AT.3a

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 **SELECTED RESPONSE — CORRECT: B**

A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

- ☐ (A) (7, 20)
- ☒ (B) (8, 24)
- ☐ (C) (8, 22)
- ☐ (D) (10, 24)

The pattern adds 2 to x and 6 to y each time. After (6, 18): $x = 6 + 2 = 8$, $y = 18 + 6 = 24$. The next point is (8, 24).

Question 2 **SELECTED RESPONSE — CORRECT: B**

When you graph equivalent ratios, the points will always form what shape?

- ☐ (A) A curved line
- ☒ (B) A straight line through the origin
- ☐ (C) A zigzag
- ☐ (D) A circle

Equivalent ratios are proportional, so their graph is always a straight line that passes through the origin (0, 0).

Question 3 SELECTED RESPONSE — CORRECT: C

A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when $x = 7$?

- ☐ A 21
- ☐ B 24
- ☒ C 28
- ☐ D 35

The ratio is $y/x = 8/2 = 4$. For every x , $y = 4x$. When $x = 7$: $y = 4 \times 7 = 28$.

Question 4 SELECTED RESPONSE — CORRECT: B

Which point would NOT be on the graph of the ratio 1:5?

- ☐ A (2, 10)
- ☒ B (3, 10)
- ☐ C (4, 20)
- ☐ D (5, 25)

For the ratio 1:5, every y-value should be 5 times the x-value. (3, 10): $5 \times 3 = 15$, not 10. So (3, 10) does not belong.

Question 5 SELECTED RESPONSE — CORRECT: D

Points from a ratio table are plotted at (1, 4), (2, 8), and (3, 12). What is the ratio of y to x for each point?

- ☐ A 1:4
- ☐ B 2:8
- ☐ C 3:4
- ☒ D 4:1

For each point: $4 \div 1 = 4$, $8 \div 2 = 4$, $12 \div 3 = 4$. The ratio of y to x is always 4:1.

Question 6

SELECTED RESPONSE — CORRECT: D

Two students graph ratio tables. Student A plots (1,2), (2,4), (3,6). Student B plots (1,3), (2,6), (3,9). Whose line is steeper?

- ☐ A Student A
- ☐ B Same steepness
- ☐ C Cannot tell
- ☒ D Student B

Student A's ratio is 1:2 (y goes up 2 for each 1 in x). Student B's ratio is 1:3 (y goes up 3 for each 1 in x). A higher rate of change means a steeper line, so Student B's is steeper.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION**Part A — correct answer: A**

A ratio table pairs 1 hour with 45 miles, 2 hours with 90 miles, and 3 hours with 135 miles. Which ordered pair belongs on the graph of this relationship?

- ☒ A (4, 180)
- ☐ B (4, 45)
- ☐ C (4, 140)
- ☐ D (180, 4)

The relationship is 45 miles per hour, so 4 hours pairs with $4 \times 45 = 180$ miles, giving the point (4, 180).

- **B:** 45 miles is the distance for ONE hour. At 4 hours it is 4×45 .
- **C:** 140 is not a multiple of 45. Four hours at 45 miles per hour gives 4×45 .
- **D:** The coordinates are swapped. Hours is the first coordinate and miles the second.

Part B — correct answer: A

What does it mean that the graph of this relationship passes through the origin (0, 0)?

- ☒ A When zero hours have passed, zero miles have been traveled.
- ☐ B The graph must be a curve.
- ☐ C The unit rate is zero.
- ☐ D The relationship only works for whole numbers.

A proportional relationship passes through the origin because zero of one quantity corresponds to zero of the other — no distance is covered in no time.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about graphing equivalent ratios:

- ☒ A The points of a proportional relationship lie on a straight line.
- ☒ B The line passes through the origin (0, 0).
- ☒ C Each ordered pair comes from a row of the ratio table.
- ☐ D The points form a curve that bends upward.
- ☒ E The unit rate can be read as the y-value when $x = 1$.

A, B, C, and E are true. D is false — equivalent ratios graph as a straight line, not a curve.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9

WRITTEN RESPONSE · 2 POINTS

A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

Model answer: She is not correct. While the first three points follow the pattern $y = 3x$ (3, 6, 9), the fourth point should be (4, 12), not (4, 15). If you plot all four points, (4, 15) would not land on the line through the others. The ratio 4:15 does not simplify to 1:3 like the others.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.